In general, a_{ij} is the element in the ith row and j^{th} column and is referred as $(i,j)^{th}$ element.

With this notation, we can express the matrix A as $A=(a_{ij})_{m\times n}$ where $i=1,2,....m$ and $j=1,2,...n$.

The total number of entries in the matrix $A=(a_{ij})_{m\times n}$ is mn.

Note

When giving the order of a matrix, you should always mention the number of rows first, followed by the number of columns.

For example,

S.No.	Matrices	Elements of the matrix	Order of the matrix
1.	$\begin{pmatrix}\sin\theta & -\cos\theta\\ \cos\theta & \sin\theta\end{pmatrix}$	$a_{11}=\sin\theta,\ a_{12}=-\cos\theta,$ $a_{21}=\cos\theta,\ a_{22}=\sin\theta$	2×2
2.	$\begin{pmatrix}1 & 3\\ \sqrt{2} & 5\\ \frac{1}{2} & -4\end{pmatrix}$	$a_{11}=1,\ a_{12}=3,$ $a_{21}=\sqrt{2},\ a_{22}=5,$ $a_{31}=\frac{1}{2},\ a_{32}=-4$	3×2

Activity 4

(i) Take calendar sheets of a particular month in a particular year.

(ii) Construct matrices from the dates of the calendar sheet.

(iii) Write down the number of possible matrices of orders $2\times 2,\ 3\times 2,\ 2\times 3,\ 3\times 3,\ 4\times 3$, etc.

(iv) Find the maximum possible order of a matrix that you can create from the given calendar sheet.

(v) Mention the use of matrices to organize information from daily life situations.



3.9.2 Types of Matrices

In this section, we shall define certain types of matrices.

1. Row Matrix

A matrix is said to be a row matrix if it has only one row and any number of columns. A row matrix is also called as a row vector.

For example, $A=\begin{pmatrix}8 & 9 & 4 & 3\end{pmatrix}$, $B=\begin{pmatrix}-\frac{\sqrt{3}}{2} & 1 & \sqrt{3}\end{pmatrix}$ are row matrices of order 1×4 and 1×3 respectively.

In general $A=\begin{pmatrix}a_{11} & a_{12} & a_{13} & \ldots & a_{1n}\end{pmatrix}$ is a row matrix of order $1\times n$.

2. Column Matrix

A matrix is said to be a column matrix if it has only one column and any number of rows. It is also called as a column vector.

10th_Maths_Chapter 3_English.indd 139 12/23/2024 5:34:17 PM